

BlitzBook

GST Invoice & Accounts · Build documentation: Android app, web portal, backend

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Source commit	91ef19d on master (44 commits), github.com/tutorialtrack1-eng/vanyagstinvoice
Android app	com.vanyaliving.invoice, version 1.0 (versionCode 1), debug APK 18.9 MB
Web portal	webportal/ (plain HTML, CSS, JavaScript, no build step)
Backends	Supabase project cufdskrmhdenppoxfhnk (default), or the Node sync server <code>server/server.js</code>
Domains	blitzbook.co.in (portal), api.blitzbook.co.in (Node server on Render)

This document describes what is actually in the code at the commit above. Where the README files in the repository say something different, the code is described and the README is flagged in section 12.

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1. Build summary and components

Component	Technology	Build / output
Android app	Java, Android SDK only. minSdk 29, targetSdk 35, compileSdk 35. AndroidX appcompat 1.7.0, material 1.12.0, drawerlayout 1.2.0. Android Gradle Plugin 9.4.1, Gradle 9.6.0. Every screen is built in code (no layout XML).	<code>gradlew assembleDebug</code> (or Android Studio, Build > Build APK). Output <code>app/build/outputs/apk/debug/app-debug.apk</code> . A copy is published as <code>webportal/BlitzBook.apk</code> for the portal's Download App link. No release signing config exists yet.
Web portal	<code>index.html</code> plus <code>css/app.css</code> and 12 JavaScript files (<code>util</code> , <code>store</code> , <code>subscription</code> , <code>appformat</code> , <code>supabase</code> , <code>sync</code> , <code>print</code> , <code>app</code> , <code>invoice</code> , <code>ledger</code> , <code>money</code> , <code>reports</code>). Google Fonts "Plus Jakarta Sans", optional.	No build. Open <code>index.html</code> as a file, serve the folder from any static host, or let <code>server.js</code> serve it. The Docker image bundles it.
Supabase backend	Supabase Auth (accounts, OTP emails), Postgres tables <code>books</code> , <code>profiles</code> , <code>activation_codes</code> , functions <code>identity_login</code> , <code>redeem_code</code> , trigger <code>profile_sync</code> . Row-level security.	<code>server/supabase/schema.sql</code> then <code>activation.sql</code> , run in the SQL Editor. Project URL and publishable key are built into the app and the portal.
Node sync server	<code>server/server.js</code> and <code>notify.js</code> , Node 18+, no dependencies. Accounts, OTP by SMTP or SMS, record sync, static hosting of the portal.	<code>node server/server.js</code> . Docker (<code>node:20-alpine</code>) with Caddy, systemd unit, Windows scheduled task, or Render blueprint (<code>render.yaml</code>).
Tools	<code>tools/LicenceKeyGen.java</code> (offline activation codes), <code>webportal/tools/smoke.js</code> (browser test), <code>server/test.js</code> and <code>server/supabase/test.js</code> .	See section 12.

Android manifest

- App name "BlitzBook", toolbar title "BLITZBOOK". Theme Material Light without action bar; accent #1E5AA8; each launch picks a random pastel tint for cards.
- Permission: only INTERNET (IFSC lookup, sync). PDFs and backups go to Downloads through MediaStore, so no storage permission.
- Declares queries for UPI links, https links and SMS, used by the subscription payment flow.
- Activities, all portrait: LoginActivity (launcher), RegisterActivity, MainActivity. Cleartext traffic is allowed for an http sync server on the LAN.

Built-in constants

Supabase URL	<code>https://cufdskrmhdenppoxfhnk.supabase.co</code> (<code>Supabase.java</code> and <code>webportal/js/sync.js</code>)
Supabase key	Publishable key <code>sb_publishable_...</code> built into both sides; a different key or server can be typed under Sync
Vendor UPI	<code>blitzbook@upi</code> , payee "BlitzBook", vendor phone 8074386833 (WhatsApp and SMS for activation requests)
Offline code secret	VANYA-INVOICE-BOOK-2026 (<code>Subscription.java</code> and <code>subscription.js</code> , used by <code>LicenceKeyGen</code>)

Activation server	Blank. After payment the request is sent by hand on WhatsApp or SMS.
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2. How the pieces fit together

BlitzBook is one set of books that can be worked on from an Android phone and from a browser. The app and the portal hold the same record types in the same table layout, and either of two backends keeps them in step:

- **Supabase** (default, nothing to run): Supabase Auth holds the accounts and sends the registration and password-reset OTPs by email. One table, `books`, holds every record of every account as a JSON row keyed by record type and id. Activation codes are redeemed through the `redeem_code` function.
- **Node sync server** (self-hosted): the same requests served by `server.js`, with accounts in `accounts.json` and one `books` file per account. It can also send OTPs by SMS and email and serves the portal itself.
- **No backend**: both sides work on their own, and data moves between them by backup file (Export / Import on either side).

Which backend is used is decided by the address under **Sync**: a `*.supabase.co` address with a key means Supabase, any other address means a Node server, and the built-in default is the Supabase project. The portal, when served from localhost or a private LAN address, uses the server it was opened from.

What is shared when sync is on

- The account itself: register on one side, log in on the other.
- Company profile with the signature image, item master, contacts, invoices, purchases and quotations, expenses, credit and debit notes, journal vouchers (including receipts and payments), ledger accounts.
- The free-period start, the paid validity and the used activation codes: one free period and one activation per account, whichever device it is used on.

Not shared: the login table on the phone (local accounts), and the app's PDF history table.

Sync mechanics in one paragraph

Every record travels as a copy of the app's table row under a key that means the same record everywhere (`inv:<invoice no>`, `item:<name>`, `contact:<id>` and so on). A client keeps a fingerprint of each record as the server last saw it, sends the ones that changed, and receives everything newer than the revision it last saw. The app runs a round within about two seconds of any write and otherwise every 10 seconds while it is in the foreground; the portal pushes 1.2 seconds after a save and polls every 10 seconds while the tab is visible. When two devices change the same record before syncing, the one that syncs last wins. The subscription record is merged instead: earliest start, latest validity, every used code. The first contact with a server only listens, so a fresh browser receives the books before showing the company dialog.

3. Accounts: register, login, password reset

The same flow exists in the app and the portal, with the same labels.

Register ("Create Account")

- Subtitle: "Register and use BlitzBook free for 30 days".
- Fields: Full Name (required), Mobile Number (10 digits, must start 6 to 9), Email (optional in the portal; "Email Address (optional)" in the app), Password (at least 6 characters with a letter, a digit and a special character).
- Buttons: Send OTP, then Enter 6-digit OTP, Verify & Register, Resend OTP.
- OTP delivery: by email from Supabase (or SMS when an SMS provider is configured in Supabase); by SMS and email from the Node server when SMTP or SMS settings are given. Without either, test mode shows the OTP on screen: "(Test mode) Your 6-digit OTP is:".
- Checks: the number or email must not have changed since the OTP was sent; a number or email already registered is refused ("This mobile number or email already has a BlitzBook account. Please log in."); the OTP is checked by the backend, or locally in test mode.
- On success the account is created locally and on the backend, the free period starts, and the message reads "Registered. BlitzBook is activated for 30 days." (app toast) or "Welcome, <name>! BlitzBook is activated for 30 days." (portal, green toast that fades out). The app dashboard then shows a one-time green note "Activated for 30 days · Activated till dd/mm/yyyy (30 days left)" that fades away after five seconds.

Login

- Fields: "Mobile Number / Email" and "Password". Links: "Forgot / Reset Password?", "New User? Register Now", "Sync settings".
- Order: the local account list first, then the backend ("Signing in..."). An account made on another device is fetched from the backend and saved locally. Offline with no local copy: "Connect to the internet to log in to an account made on another device."
- Passwords never reach a server as typed: the client sends SHA-256("bb|" + password). The Node server stores that under script; Supabase stores it as the Auth password.
- The session is remembered; the app and the portal open straight to the dashboard next time. Logout asks "Do you want to logout?".
- If the password is changed on another device, the next sync gets 401 and the device is signed out with "Log in with the new password."

Forgot / Reset Password

- Step 1: "Registered Mobile / Email", "Send Reset OTP". The OTP goes to the account's mobile number and email (Node server), or by email from Supabase.
- Step 2: OTP, New Password, Confirm New Password, "Reset Password". A successful reset signs every other device out.
- A device that is signed out of the account and offline cannot reset: "This phone is signed out of the account. Log in with the current password, or reset it on a device that is signed in."
- Admin alternative on the Node server: `node server/server.js passwd <mobile-or-email> <new-password> .`

Sync settings

Reachable from the login screen, the app's menu ("Sync") and the portal's Export / Import screen. Fields: "Sync server address or Supabase project URL" and "Supabase anon key (Supabase projects only)". Buttons Sync Now (signed in) or Check (signed out). The status line reads Synced, Syncing..., Offline, Not connected, Sign in again, or This device only / This phone only.

4. Free period, subscription and activation codes

Free 30 days

- Every new account may use BlitzBook for 30 days from the first time it opens the app or portal. Status text: "Activated till dd/mm/yyyy (N days left)".
- The Company Master Profile ends with an **Activation** card: the status plus "N days remaining", green while active and red once expired, with the hint "Activation codes are entered under Subscription in the menu / top bar".
- When the period runs out, a non-cancellable "Subscription Required" dialog appears: "Your free 30-day activation has ended. A subscription is needed to continue." with an Activation Code field and the buttons Activate, Buy / Renew and Logout. The same dialog opens from the menu as "Subscription" with Close instead of Logout.

Plans and payment

Plan	Price
1 day	Rs 49
30 days	Rs 299
90 days	Rs 799
180 days	Rs 1,499
365 days	Rs 2,499
730 days	Rs 3,999

- "Buy / Renew" opens "Choose a Plan", then a UPI payment: the app opens any UPI app with the amount and the note "BlitzBook <days>d <phone>"; the portal shows the payee, the note, an "Open UPI app" link and a "Transaction Reference" field.
- After "I have paid" the request is recorded as pending ("Payment recorded: Plan X days, Rs N, UPI ref ...", waiting for the activation code on ...) and shown in the Subscription dialog until a code is entered. The request can be sent on WhatsApp or by SMS to 8074386833 with the details prefilled.

Activation codes

- A code is 16 letters and digits, typed with or without dashes (XXXX-XXXX-XXXX-XXXX).
- **Codes issued in Supabase** (the normal way): rows in the `activation_codes` table, each worth a number of days. A code works once, on any account. The app and the portal call `redeem_code`, which checks and marks the code for the signed-in account. The table is not readable by clients, and the Supabase Table Editor shows which codes are used and by whom. The first 20 codes (5 × 1 month, 5 × 3 months, 4 × 6 months, 4 × 1 year, 2 × 2 years) are loaded by `activation.sql` and listed in `server/supabase/activation-codes.txt`; more are issued by inserting rows.
- **Offline codes** (fallback, no backend needed): generated with `java tools/LicenceKeyGen.java <mobile-or-email> <days>`, tied to the login identity and a plan length through SHA-256 and the shared secret.
- The validity starts the moment the code is entered and runs for the plan's days. Days left from before are not added. Status: "Subscription valid till dd/mm/yyyy (N days)".
- Errors: "This code has already been used", "Invalid activation code". Used codes are remembered per account and synced, so a code cannot be entered twice on another device.

5. Feature matrix: app versus web portal

Feature	Android app	Web portal	Notes
Register, login, reset password, logout	Yes	Yes	Same account on both sides with sync on
Free 30 days, subscription dialog, UPI plans, activation codes	Yes	Yes	Shared through the sub record
Dashboard: greeting, company chips, month stats, recent products, tiles	Yes	Yes	Portal adds New Invoice / View Sales buttons and animated counters
Company Master Profile with GSTIN check, IFSC lookup, signature, activation card	Yes	Yes	Invoice Number Format field is in the portal only; the app reads the format once it syncs
Invoice editor with GST, RCM, consignee, other details, sub-details, quick picker	Yes	Yes	Portal asks before opening an existing number; app loads it
Print: standard and classic layouts, paper sizes, envelopes, delivery challan, e-way bill warning	Yes	Yes	App writes PDF files to Downloads/BlitzBook; portal prints through the browser (Save as PDF)
Sales register, open / print / delete	Yes	Yes	
Credit notes and debit notes with caps against the document	Yes	Yes	
Customers and suppliers, TDS, CSV / Excel upload, template, bulk delete	Yes	Yes	
Item master, bulk category / GST / delete	Yes	Yes	Dashboard "Stock" tile opens the item master on both sides
Purchases, quotations, convert to purchase, RCM, inclusive rates, TDS, stock flag	Yes	Yes	
Stock in hand, stock upload, delete with or without the master item	Yes	Yes	
Expenses with GST (bill value or taxable), vendor GSTIN, RCM	Yes	Yes	
Journal vouchers with auto-balance and account picker	Yes	Yes	
Receipts & Payments editor, outstanding balance, open bills, voucher print	Display only	Yes	The app lists them in the Journal and uses them in the books; no editor in the app
Bank statement upload with party matching and duplicate detection	No	Yes	
Sales report, Profit & Loss, Balance sheet, Excel export	Yes	Yes	Portal adds CSV export of the sales report
Export / Import backup in the app's table layout	Yes	Yes	A backup from one side restores on the other

Feature	Android app	Web portal	Notes
Sync settings	Yes	Yes	Portal: under Export / Import and on the login screen
Download App link	n/a	Yes	Top bar, serves BlitzBook.apk
Share invoice PDF (share sheet)	Yes	Browser print	

6. Android app: every screen and function

Start-up

- A remembered session goes straight to the dashboard. The free-period clock starts on the first open. Database columns and ledger tables are created or migrated.
- With no company profile the Company Master Profile opens. With sync on and nothing synced yet it first waits for the books ("Checking for your books...").
- Sync runs while the app is in the foreground and stops on pause. The subscription is checked on every resume and on a timer.

Dashboard

- Banner: "Good morning / afternoon / evening · <date>", the company name (or "Set up your Company Profile"), chips "GSTIN ..." or "No GSTIN" and the line of activity.
- One-time green note after registration: "Activated for 30 days · ...", fading out after five seconds.
- "Recent products": up to 8 item chips from the latest invoices; tapping one starts an invoice with that item.
- "What would you like to do?": Invoice, Sales, Customer, Supplier, Purchase, Stock (item master), Expense, Journal, Reports (menu: Sales Report, Profit & Loss, Balance Sheet, Stock in Hand).
- "This month" cards: Sales this month (rounded totals from the 1st at midnight), Invoices (count), Credit outstanding (all invoices paid by Credit).

Menu (≡ drawer)

Header with "BlitzBook" and the company name; tiles Company Profile, Sales Report, Profit & Loss, Balance Sheet, Stock in Hand, Export / Import, Subscription, Sync; a "Sync: <status>" line; a red Logout button.

Company Master Profile

Card	Fields and behaviour
Company & Contact Details	Company Name*, GSTIN* (15 characters, live check of the mod-36 checksum and state code 01 to 38), GST Registration Type* (Regular / Composition / Unregistered; clearing the GSTIN switches to Unregistered, typing one switches to Regular), Line of Activity (Food and Beverages, Retailer, Services, Manufacturing, Wholesale, General), Address*, Phone (10 digits)*, Email* (phone and email start as the registered account's).
Bank Account Details	Account Number, Account Holder Name (capitals, defaults to the company name), IFSC Code (AAAA0XXXXXX; at 11 characters the bank name fills from a built-in table of about 40 banks and bank plus branch are looked up at ifsc.razorpay.com), Bank Name, Branch Name. Messages: "Looking up bank & branch...", "✓ Bank & branch filled from IFSC", "IFSC not found. Please check the code.", "Could not reach IFSC lookup. Enter bank & branch manually."
Authorised Signature	Attach Signature (image picker, scaled to 600 px wide, saved as PNG) and Remove. Prints above "Authorised Signatory".
Activation	Status and days remaining, green or red; hint on where codes are entered.

Save checks: name, address, phone pattern, email, GSTIN; "GSTIN given: select Regular or Composition"; "GSTIN is required for <type> dealer"; IFSC format. Address is stored in capitals. Toast "Company Profile Saved Successfully!".

What the GST type changes: only Regular charges GST and prints "TAX INVOICE"; Composition hides every GST field, prints "BILL OF SUPPLY" and gets a Delivery Challan button; Unregistered prints "INVOICE". The reverse-charge option on sales appears only for Regular dealers whose activity is Services.

Invoice

- **Invoice Details:** Invoice No* with ▲/▼ (next number = highest saved number in the format plus one; the format may contain {FY}, for example INV/{FY}/#####, so numbering restarts each financial year; typing an existing number loads that invoice), Dated* (date picker, long-press clears), Payment Mode (Cash / Online / Cheque / Credit), "Reverse charge (RCM) - GST payable by the recipient" for service businesses.
- **Buyer & Shipping Details:** Buyer (Bill To)* with autocomplete from contacts (fills address, phone, email, GSTIN, state), Phone, Email, GSTIN, State (36 states and UTs with codes); "Shipping same as Billing" (on by default) or a separate Consignee (Ship To) block.
- **Other Details:** Destination, Vehicle Type, Vehicle Number; "Show additional details" reveals Transporter, Delivery Note, Buyer Order No, Buyer Order Date, Reference No, Other Info.
- **Goods / Services:** a "Quick Items / Food Items / Services / Products" button (label follows the activity) and the item table: Sl, Particulars (autocomplete from the item master, "+" for Product Sub-Details: Serial No., Description, Other Info / Warranty), HSN/SAC (filled from the master or the built-in HSN map), GST % (0, 5, 18, 40, 3, 0.25), Inc? (rate includes GST: type the Total Incl. and taxable and rate are backed out), Qty*, UQC (33 codes), Rate*, Taxable / Amount, Total Incl., delete. "+ Add Particular / Row" needs the current row complete.
- **Totals Summary:** Taxable Value, CGST, SGST, IGST, Grand Total, Rounded Total, Amount in Words (Indian style, "INR ... only").
- **Buttons:** SAVE ("Invoice X saved"), PRINT / PDF, new, delete (blocked while credit notes exist), DELIVERY CHALLAN (Composition), Layout & Paper.
- **Validation:** date required; buyer required for Credit or Cheque ("Buyer details are mandatory for Credit sales"); phone, email and GSTIN patterns for buyer and consignee; every row needs particulars, quantity and rate ("Please add at least one item").
- **Saving** replaces any invoice with the same number and upserts each item into the item master. **Print / PDF** warns about the e-way bill above Rs 50,000 (Rs 1,00,000 in Maharashtra, Delhi, Tamil Nadu and Bihar), saves, writes Downloads/BlitzBook/<no>.pdf and offers "Print / Share PDF", "Next Invoice (Just Save)" and "Stay Here".

Quick items picker (" ⚡ Quick ...")

Starter items for the line of activity merged with the item master; search by name or code; list or grid view; category chips (long-press to rename or remove a category, items move to Catalog); a -/qty/+ stepper per item; "₹ x · GST y% ↘" for a per-invoice price and GST or "Customise Item"; "+ Add" opens the item editor (Item Name*, Item Code, Category, HSN/SAC, GST Rate %, Unit Price; Remove hides the item). Summary "Selected: N items | Total: ₹ x"; "Apply to Invoice" adds, updates or removes rows.

Sales

Buttons "+ New Invoice", "Credit Notes", "Report". Every invoice newest first with number, date, buyer or " (cash sale)", total, mode and RCM; icons Open, Print, Delete.

Credit notes and debit notes

Numbered CN-0001 / DN-0001. Fields: Note No, Date*, Customer* or Supplier*, Against Invoice* or Against Purchase* (must exist), Party GSTIN, Reason, Taxable Value* (capped at the document's taxable value less earlier notes), GST Rate %, Settlement (Credit / Cash / Online / Cheque). GST splits into IGST or CGST plus

SGST by the party's state. Each note can be printed (A4), edited and deleted. Debit notes are reached from Purchases.

Customers and suppliers

" + Add", "Upload CSV" (CSV or .xlsx; an existing name is updated, "N contacts added, M updated"), "Template" (Name, Phone, Email, GSTIN, Address, State), "Select" (Select all, Delete selected). Editor: Contact Type*, Name / Company Name*, Phone, Email, GSTIN, State, Address, "TDS applicable" with TDS Section (194C, 194J, 194H, 194I, 194Q, 194A, Other) and TDS Rate % (above 0, at most 30).

Purchases & Quotations

Buttons "+ Purchase", "+ Quotation", "Stock", "Debit Notes". Numbers PUR-0001, QTN-0001, STK-0001 (stock uploads). Rows show number, date, kind, supplier, total, mode, RCM GST, TDS and stock items; quotations have "To Purchase"; every row prints (A4 / A5 / Letter / Legal), edits and deletes. Editor: number, Date*, Supplier (fills GSTIN), Supplier GSTIN, Paid By or Expected Payment, "Reverse charge (RCM) - GST payable by us, not to the supplier", "Item rates include GST", "Deduct TDS @ r% (section)" when the supplier has TDS; item rows with Item, HSN, Qty, Rate, GST %, Stock; live Taxable, GST, Total or Bill value, Less TDS, Payable to supplier; Notes. Items ticked as stock join the item master.

Stock in Hand

"Upload Stock" (Item, HSN, Qty, UQC, Rate, GST%) and "Template". Columns Item, Bought, Sold, In Hand (red at zero or below), Rate, Value. Bought counts stock-flagged purchase lines and uploads, Sold counts invoice lines by item name, value is on hand at the last purchase rate. Total "Stock value (at last purchase rate)".

Item Master (dashboard "Stock" tile)

" + Add Item", "Select" with Select all, Set category, Set GST %, Delete. Rows show code, name, HSN, GST, price and category with edit and delete. Invoiced and purchased items appear automatically.

Expenses

" + Add Expense"; header "N entries | Total ₹x". Editor: Date*, Paid By, Category* (Rent, Salaries & Wages, Electricity & Water, Transport & Fuel, Office Supplies, Marketing & Advertising, Repairs & Maintenance, Professional Fees, Bank Charges, Telephone & Internet, Packing Material, Insurance, Taxes & Licences, Other), for GST dealers "Enter bill value (incl. GST)" or "Enter taxable value + GST" with Bill Value, GST Rate %, Taxable Value, GST Amount, Vendor GSTIN and reverse charge; otherwise Amount*; Description.

Journal Entries

" + Add Journal Entry". Date*, any number of Dr / Cr lines with an account picker and amount, "+ Debit line", "+ Credit line", auto-balancing of the last untyped line, status "Debit x Credit y (balanced)" or "Difference", Narration. Account picker: "+ Create new party / account..." (Customer, Supplier, Expense, Income, Asset, Liability, Bank, Cash / petty cash), built-in accounts (Cash, Bank, Capital, Drawings, Sales, Other Income, Purchases, Depreciation, Fixed Assets, Loans, Output and Input CGST / SGST / IGST, RCM GST Payable, TDS Payable), every contact, user accounts. Receipts and payments made in the portal are listed here as "Receipt / Payment <no> from / to <party> against <ref>".

Reports, Export / Import, Subscription, Sync

Reports are described in section 9 (same figures as the portal), Export / Import in section 10, Subscription in section 4 and Sync in section 3. Report exports are written to Downloads/BlitzBook as BlitzBook_Sales_Report, Profit_Loss or Balance_Sheet files.

7. Web portal: every screen and function

Opening it

- Open `webportal/index.html` directly, serve the folder from any static host (blitzbook.co.in), or run `node server/server.js` and open `http://localhost:8080/`.
- Data lives in the browser's localStorage under the prefix `blitzbook.`, per account (`blitzbook.u<id>.<collection>`), plus the account list, session and sync settings. With a backend the books are kept there too.
- Responsive from phone to desktop: tables turn into cards on narrow screens, the navigation strip scrolls, the active screen is marked.

Top bar

- Left: the BlitzBook brand (goes to the dashboard) and the navigation: Dashboard, Company Profile, Sales Report, Profit & Loss, Balance Sheet, Receipts & Payments, Stock in Hand, Export / Import, Subscription.
- Right: **Download App** (BlitzBook.apk), the company chip (avatar letter and company name, opens the profile), Logout.
- Inner screens carry "◀ Back to Dashboard".

Dashboard

- Hero: greeting with the date, company name, chips GSTIN and activity, buttons "New Invoice" and "View Sales".
- Stats with counting animation: "Sales this month", "Invoices" ("Raised this month"), "Credit outstanding" ("Across all credit invoices").
- "Recent products" chips and the tiles Invoice, Sales, Customer, Supplier, Purchase, Stock, Expense, Receipts, Journal, Reports (titles only, no descriptions).
- First login with no company name opens the Company Master Profile in first-time mode ("Later" / "Save Profile"); a fresh browser waits for the first sync first.

Company Master Profile

The same fields and checks as the app (section 6), plus **Invoice Number Format** ("# = digits, {FY} = financial year, e.g. INV/{FY}/{#####}"). The signature is stored as a PNG data URL and synced to the app. The dialog ends with the Activation card and the days remaining.

Invoice editor

Identical fields, calculations and validation to the app (section 6), with these portal specifics:

- Typing an existing number asks "Invoice X already exists. Open it?".
- A GSTIN typed for the buyer or consignee sets the state from its first two digits.
- Buttons: Delete, + New, Print Settings, Delivery Challan (Composition), Save, Print / PDF. Save moves straight to the next number ("Invoice X saved. Next invoice: Y"). Print / PDF saves, prints with the chosen layout and offers "Stay Here", "Next Invoice", "Print again".
- Print Settings: live previews of the two layouts ("In use" / "Use this layout"), paper A4 / A5 / Letter / Legal, and "Envelope (addresses only)" DL / C5 / #10 printed straight away. Layout and paper are saved in the company record.

- The buyer's first line becomes a Customer contact when new.
- Quick picker with the same search, chips, stepper, per-invoice price and item editor as the app; new names are name-cased and duplicates refused.

Sales, notes, contacts, items, purchases, stock, expenses, journal

The same screens, labels, numbering (CN, DN, PUR, QTN, STK) and rules as the app (section 6). Portal extras: the Sales list has a "Receipts" button; Purchases has "Payments"; the item master has a "Stock in Hand" button and Stock in Hand an "Item Master" button; the Journal has "Receipts & Payments"; stock delete asks "Also delete from the item master (no longer offered on invoices)". Contact upload accepts an optional Type column.

Receipts & Payments (portal only)

- Header: "+ Receipt", "+ Payment", "Upload Bank Statement", "Template". Filter chips All / Receipts / Payments; totals "Received ₹... | Paid ₹...". List columns Date, No, Type, Party / Account, Against, Mode / Ref, Amount with Print, Edit, Delete.
- Numbers RCT-0001 / PMT-0001. Each voucher is a journal entry: a receipt debits Cash or Bank and credits the party; a payment debits the party and credits Cash or Bank. Cash mode uses Cash; everything else uses Bank.
- Editor: number, Date*, "Received from" / "Paid to"* (Customers, Suppliers, Accounts, "+ New party / account..."), an outstanding hint ("Outstanding: ₹... owed to you / owed by you" or "No outstanding balance"), Amount*, "Received in" / "Paid from" (Cash, Bank Transfer, UPI, Cheque, Card), "Against invoice" or "Against purchase / bill" (the party's open credit bills; choosing one fills the amount), "Bank / UPI / Cheque ref", Narration.
- Receipt and payment vouchers print on A4 (section 8).

Bank statement upload (portal only)

- Accepts CSV or .xlsx from any bank, or the template (Date, Description, Debit, Credit, Reference). The header row is found by column names (date, narration / description / particulars, withdrawal / debit and deposit / credit, or amount with Dr / Cr) within the first 40 rows.
- Dates in dd/mm/yyyy, dd-mm-yy, dd.mm.yyyy, dd-MMM-yy, dd MMM yyyy, MMM dd yyyy, yyyy-mm-dd and Excel serials; amounts strip ₹, commas, INR, Rs, Cr / Dr; parentheses mean negative.
- Money in becomes a receipt, money out a payment. The party is guessed from earlier vouchers with the same narration (learnt and remembered), else from a customer or supplier name found in the narration.
- Lines already recorded (same date, amount and narration) are greyed as "Already recorded" and unticked. The dialog "Bank Statement: N transactions" has Select all, a summary, a party select and an "Against" field per line. "Record" creates the vouchers with mode Bank Transfer: "N transactions recorded, M skipped (no party chosen)".

Export / Import

"Export Backup" (BlitzBook_Backup_<stamp>.json), "Import Backup" (confirmation, then a file picker; result "Restored N invoices, M items and K contacts..."), "Sync settings" and the sync status line. The file is the app's table layout, so it restores in the app and vice versa; older portal files and the old two-line journal format are converted on import.

8. Printing and documents

- Titles: TAX INVOICE (Regular), BILL OF SUPPLY (Composition), INVOICE (Unregistered), DELIVERY CHALLAN, QUOTATION, PURCHASE RECORD, CREDIT NOTE, DEBIT NOTE, RECEIPT, PAYMENT VOUCHER.
- Paper: A4 (default), A5, Letter, Legal. Envelopes DL 220 × 110 mm, C5 229 × 162 mm, #10 9.5 × 4.125 in.
- Every printout ends with "Powered by BlitzBook"; invoices add "Page x of y" (app) and repeat the table heading on every page (portal).
- App: PDFs are written to Downloads/BlitzBook (invoice number, DC_, Envelope_ prefixes) and shared with the Android share sheet. Portal: the document is printed through the browser's print dialog with zero page margins so no browser header or footer appears; "Save as PDF" makes the file.

Standard (BlitzBook) layout

Underlined title; seller name, address, GSTIN line ("(Composition Dealer)" or "Not Registered under GST"), phone and email; Invoice No, Date, Payment, Reverse Charge; a three-column BILL TO / SHIP TO / OTHER DETAILS block (Dest, Veh No, Trnsp, Challan, Ord No, Ord Dt, Ref, Info); items with SI, PARTICULARS and sub-details, HSN/SAC, GST RATE, Qty, Rate, Amount; "GST Breakdown" per rate (CGST and SGST halves, or IGST); Amount in Words; the RCM declaration (Sec 9(3)/9(4)) when applicable; BANK DETAILS box; totals with Taxable, taxes, Grand Total and Rounding; "For <company>", the signature image, "Authorised Signatory", or "Computer-generated document. No signature required."

Classic boxed (Tally style) layout

A bordered frame with the seller, "Consignee (Ship to)" and "Buyer (Bill to)" on the left (GSTIN/UIN, State Name and code, E-Mail, Phone), and a grid on the right: Invoice No., Dated, Delivery Note, Mode/Terms of Payment, Reference No. & Date, Other References, Buyer's Order No., Dated, Dispatched through, Destination, Motor Vehicle No., Reverse Charge. Items with SI No., Description of Goods, HSN/SAC, GST Rate, Quantity, Rate, per, Amount; tax rows, Round Off and a Total row with the quantity sum under the Quantity and Amount columns; "Amount Chargeable (in words)" with "E. & O.E."; an HSN/SAC-wise tax table and "Tax Amount (in words)"; "Company's Bank Details", the Declaration, signature, "Authorised Signatory", "This is a Computer Generated Invoice".

Other documents

- **Delivery Challan** (Composition dealers): DESCRIPTION OF GOODS, HSN/SAC, Qty, UQC, Rate, Value, total quantity, "Value of Goods" in words, the note that it is not a bill of supply or tax invoice, Receiver's Signature box.
- **Envelope**: "From:" block with the seller's address, phone and GSTIN at top left; "To," with the buyer's name, address, state, phone and GSTIN at lower right; "Ref: Invoice X dated Y".
- **Quotation / Purchase Record**: supplier box with GSTIN and notes; items with SI, DESCRIPTION ("(stock)"), HSN/SAC, Qty, Rate, GST %, Amount; Taxable, taxes, Total or "Payable to Supplier" under RCM; words; "valid for 30 days" on quotations; signature.
- **Credit / Debit Note** (A4): Note No, Date, Against; ISSUED TO with GSTIN and Reason; particulars with GST and amount; TOTAL; words; settlement text ("Adjusted against the customer's account" or "Refunded by <mode>").
- **Receipt / Payment voucher** (A4, portal): number, date, against invoice or bill; RECEIVED FROM or PAID TO with narration; "Amount received / paid by <mode> (ref ...) against ..."; total in words in rupees and paise; "Received with thanks. Subject to realisation of cheque / transfer where applicable." on receipts.

9. Accounting rules, reports and calculations

GST on a document

- Place of supply: intra-state when the buyer's state code equals the first two digits of the seller's GSTIN (37, Andhra Pradesh, when there is no GSTIN). Intra-state: $\text{CGST} = \text{SGST} = \text{taxable} \times \text{rate} / 2$. Inter-state: $\text{IGST} = \text{taxable} \times \text{rate}$.
- Inclusive rates: $\text{taxable} = \text{total} \div (1 + \text{rate} / 100)$; the rate per unit is backed out from the quantity.
- Reverse charge on a sale: the invoice total is the taxable value and the tax rows read "(RCM, payable by buyer)"; on a purchase or expense the GST is "paid by us" and appears as RCM GST payable, not as part of the supplier's bill.
- Rounded total = nearest rupee; the amount in words is in lakh and crore form, "INR ... only."; tax amounts in words carry rupees and paise.
- TDS on a purchase = $\text{taxable} \times \text{the supplier's rate}$, rounded to paise; "Payable to supplier" = total less TDS; the TDS appears as TDS payable.
- E-way bill reminder above Rs 50,000 (Rs 1,00,000 for Maharashtra, Delhi, Tamil Nadu and Bihar).

Period picker

This Month, Last Month, This Quarter, This Financial Year (April to March), Financial Year (choose previous, current or next), Custom Range (From / To, Generate).

Sales Report - Invoice Wise

Summary of Period, Invoices, Taxable, GST (zero on RCM invoices), Grand Total; a table of Invoice No, Date, Buyer Name, Taxable Value, GST Amount, Grand Total; "Export Excel" (and CSV in the portal). The Excel file is an HTML table saved with the .xls extension, which Excel opens directly.

Profit & Loss

- INCOME: Sales before GST, less credit notes, other income from the journal.
- COST OF GOODS: purchases before GST (purchases only, not quotations or stock uploads) plus journal purchase accounts, less debit notes. Gross Profit.
- EXPENSES: by category at taxable value, plus journal expense accounts. Total Expenses, then NET PROFIT or NET LOSS.
- For Regular dealers a "GST (not part of profit)" block: Output CGST / SGST / IGST (RCM sales excluded), Input CGST / SGST / IGST, RCM GST payable, "Net GST payable" or "Net GST credit".

Balance Sheet (as at a date)

- ASSETS: Cash in hand; Bank (online and cheque); Receivables with each outstanding customer listed; Stock in hand at the last purchase rate; asset accounts; Input GST; Total Assets.
- LIABILITIES & CAPITAL: Payables with each outstanding supplier listed; liability accounts; Output GST and the net GST line; GST payable under reverse charge; TDS payable; Owner's capital (accumulated profit or loss, the balancing figure); Total.
- Mode logic: Cash goes to cash; Credit goes to the party (receivable or payable); Online, Cheque, UPI and Card go to bank. Receipts and payments, credit and debit notes net off per party through the journal.

Stock in hand

Bought = quantities on stock-flagged purchase lines and stock uploads; Sold = invoice quantities by item name; In Hand = bought less sold; Value = in hand (not below zero) × last purchase rate.

10. Data storage and sync

Android app storage

- `vanya.db` holds the login table (users: id, name, email, password, phone) and the books of the first (legacy) owner; every other account has its own `invoicebook_user_<id>.db`.
- Tables: `company_master`, `items_master`, `contacts`, `invoices`, `invoice_items`, `expenses`, `purchases`, `purchase_items`, `journal_vouchers`, `journal_lines`, `ledger_accounts`, `notes`, `history` (PDF log, not synced), `sync_state`, `sync_meta`. The old journal table is migrated into vouchers.
- SharedPreferences "invoice_prefs": session (is_logged_in, user_id, user_email), subscription (registered_at, valid_until, used_codes, pending_activation, welcome, per account), sync (sync_server_url, supabase_key, sync_token per account), printing (pdf_layout, pdf_paper).
- Signature: `signature.png` or `signature_<id>.png` in the app's files directory. Dates are stored as dd/MM/yyyy text.

Portal storage

localStorage keys `blitzbook.users`, `blitzbook.session`, `blitzbook.sync_url`, `blitzbook.supabase_key`, and per account `blitzbook.u<id>`. `contacts`, `items`, `invoices`, `purchases`, `expenses`, `journal`, `notes`, `accounts`, `company`, `registered_at`, `valid_until`, `used_codes`, `pending_activation`, `sync`.

Backup file

One JSON file in the app's table layout: `company_master`, `items_master`, `contacts`, `history`, `invoices`, `invoice_items`, `expenses`, `purchases`, `purchase_items`, `journal_vouchers`, `journal_lines`, `ledger_accounts`, `notes`, plus a portal block (signature, layout, paper). Login accounts are not included. Import replaces each table present in the file.

Sync record keys

Key	Content
company	Company profile row with the signature as a data URL
sub	registered_at, valid_until, used_codes (merged: earliest, latest, union)
item:<name>	Item master row
acct:<name>	Ledger account
inv:<invoice no>	Invoice with its items
contact:<id>, exp:<id>, pur:<id>, note:<id>, jrn:<id>	Contact, expense, purchase with items, credit or debit note, journal voucher with lines (receipts and payments carry kind, number, party, reference, mode, bank reference)

Requests

Request	Node sync server (POST /api/<name>)	Supabase
ping	/api/ping	GET /auth/v1/settings
exists	{identity}	rpc identity_login

Request	Node sync server (POST /api/<name>)	Supabase
otp	{purpose register reset, phone, email identity}; sends SMS and email, or returns the OTP in test mode	/auth/v1/otp (register, email or SMS) or /auth/v1/recover (reset)
register	{name, phone, email, pw, otp}	/auth/v1/verify, PUT /auth/v1/user, POST /rest/v1/profiles
login	{identity, pw}; 429 after 10 failures per 10 minutes	identity_login then /auth/v1/token? grant_type=password
password / reset	New hash, other devices signed out	PUT /auth/v1/user then /auth/v1/logout? scope=others
sync	{token, epoch, since, changes}; returns {epoch, rev, changes}	Upsert into books in batches of 200, then GET books where r > since, paged by 1000
activation	n/a (offline codes)	rpc redeem_code {code_in}

11. Backend: Node sync server, Supabase, deployment

server/server.js

- Node 18 or newer, no dependencies. Environment: PORT (8080), BLITZBOOK_DATA (server/data), BLITZBOOK_WEB (../webportal). Serves the portal as static files (refusing tools/, node_modules/ and dotfiles) and the APK with its Android media type.
- Routes: /api/ping, OPTIONS with CORS, POST otp, register, login, exists, password, reset, sync. Unknown names give 404 "Unknown request"; bodies above 24 MB give 413.
- Security: scrypt over the client's SHA-256 hash with a random salt; HMAC tokens with a per-user version so a password change signs other devices out; OTPs from crypto.randomInt stored as HMACs, valid 10 minutes, 5 attempts, 5 sends per 10 minutes per device, 20 seconds between sends.
- Data: accounts.json and books/<id>.json, written atomically; up to 500 books cached in memory. Admin: `users` and `passwd` commands.

server/notify.js (OTP delivery)

Email through SMTP_HOST, SMTP_PORT (465 TLS, or 587 / 25 with STARTTLS), SMTP_USER, SMTP_PASS, SMTP_FROM, as environment variables or config.json in the data folder. SMS through SMS_PROVIDER fast2sms, msg91 (DLT template with ##OTP##), twilio, textlocal or a generic http URL. Without either the server runs in test mode and says so at start-up.

Supabase

- `schema.sql`: table books (user_id, k, d jsonb, r revision, updated_at; a trigger stamps every write), table profiles (name, phone, email, unique), row-level security "own books" and "own profile", function identity_login (finds the email or phone behind what was typed), trigger profile_sync (keeps profiles in step with confirmed Auth users).
- `activation.sql`: table activation_codes (code, days, note, created_at, used_by, used_at) with no client access, function redeem_code (returns the days, or -1 unknown, -2 used, -3 not signed in), and the first 20 codes.
- Dashboard setup (server/supabase/README.md): the Magic Link and Reset Password email templates must carry {{ .Token }}; custom SMTP (Gmail App Password) is needed for real delivery; an SMS provider is optional; the URL and publishable key go into the app, the portal, or the Sync dialog.

Deployment options

Option	What is provided
Render	render.yaml: web service blitzbook-api, Node 20, Singapore, starter plan with a 1 GB disk at /var/data, health check /api/ping, auto-deploy from master, domain api.blitzbook.co.in (GoDaddy CNAME to blitzbook-api.onrender.com).
Docker + Caddy	server/deploy/Dockerfile (node:20-alpine, portal bundled), docker-compose.yml with caddy:2 on 80 / 443, Caddyfile reverse-proxying api.blitzbook.co.in with automatic Let's Encrypt certificates.
systemd	blitzbook.service: user blitzbook, /opt/blitzbook, data in /var/lib/blitzbook, Restart=always.

Option	What is provided
Windows	install-task.ps1 [-Port 8090] creates the hidden scheduled task "BlitzBookSync" (starts at sign-in, restarts on failure, opens the firewall when run as administrator); start-hidden.ps1 and uninstall-task.ps1.
Portal hosting	Any static host; the repository CNAME is blitzbook.co.in. The Download App link expects BlitzBook.apk next to index.html.

12. Tests, known gaps and notes

Automated checks (all passing at this commit)

- `gradlew --offline lintDebug assembleDebug` : Android lint and the debug APK.
- `node server/test.js` : the Node server against the portal's own sync code. Two simulated browsers, an app-row client, OTP flows in test mode and against a fake SMTP server, conflict resolution, password change signing the other device out, backup round trips, epoch reset.
- `node server/supabase/test.js` : the portal's Supabase backend against a stand-in for Supabase Auth and PostgREST: OTP registration, login by phone or email, sync push and pull, deletes, expired-token re-login, password reset and change.
- `PLAYWRIGHT_CORE=... node webportal/tools/smoke.js` : Chromium drives the portal through every screen. Part 1 opens it as a file (30 free days, no sync or trial chip, Download App link, days remaining in the profile). Part 2 serves it through the Node server: registration, company profile with IFSC lookup, invoice numbering and HSN fill, quick picker, print settings and layouts, envelopes, expenses, TDS purchases, quotation conversion, credit-note caps, journal balancing, CSV uploads, stock, P&L, balance sheet, receipts, bank statement matching and duplicates, backup layout; then a second browser signs in to the same account and must show the same books, receive live edits and deletes, and be unlocked by an activation code entered on the first; finally a phone-width check.

Known gaps and notes found while documenting

Item	Detail
README files out of date	The root, webportal and server/deploy READMEs still describe a 1-day trial, a Sync entry and status chip in the portal's top bar, and PDFs under "Downloads/Vanya GST Invoices". The code gives 30 days, has no top-bar sync chip, and writes to Downloads/BlitzBook.
Activation codes must be loaded	The redeem function exists in the live Supabase project, but the 20 codes in activation.sql are inserted only when that file is run in the SQL Editor.
Debug APK on the portal	BlitzBook.apk is the debug build signed with the debug key. A release signing config does not exist yet.
Receipts in the app	The app has no Receipt / Payment editor; vouchers made in the portal are displayed in the Journal and used in the books.
Credit outstanding tile	The dashboard figure is the sum of all credit invoices and is not reduced by receipts or credit notes. The balance sheet receivables are.
GST rate list	The invoice GST dropdown offers 0, 5, 18, 40, 3 and 0.25 percent. Some starter items are set to 12 percent, which still calculates correctly but cannot be picked from the list.
Hosted Node server and the portal	A portal served from a public Node server does not use that server automatically; it uses the built-in Supabase default unless the address is typed under Sync. Served from localhost or a LAN address it does use the server it came from.
Accounts made offline	An account registered with no backend reachable joins the Node server on the first sync, but cannot join Supabase later because Supabase needs the OTP at registration.
Placeholders	The UPI ID blitzbook@upi and the blank activation server mean every payment is confirmed by hand on WhatsApp or SMS. Company logo and terms fields exist in the portal's data model but have no screen or print use. No discount field exists on invoices.